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A genetic algorithm-based ensemble framework for wind speed forecasting

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Wind energy has gained significant attention as a clean, renewable resource due to fossil fuels' environmental impact. Accurate wind speed forecasting is essential to address variability and intermittency challenges. Current forecasting difficulties arise from wind speed's high susceptibility to meteorological conditions. This study proposes a GA-based ensemble framework that combines forecasting models using genetic algorithms. We systematically compared 14 models: linear models (AR, ARMA), advanced neural networks (MLP, RBF), hybrid models, and ensembles. Models were evaluated using minute-by-minute data from five major Brazilian cities: Brasília, Florianópolis, Petrolina, Natal, and São Luís. Key findings include: I) Superior Performance: The proposed framework achieved MSE values from 0.0802 to 0.9020 and MAE values from 0.1970 to 0.6140 across all datasets; II) Robust Prediction: R^2 values ranged from 0.7139 to 0.8723, demonstrating strong predictive capability; III) Statistical Validation: Friedman test ($p < 0.001$) confirmed significant differences with perfect rank stability across all locations; IV) High Scalability: Runtimes ranged from 58,077.3 to 77,815.7 μs, determined by the base model combination; and V) Computational Efficiency: One-step-ahead forecasting requires only 0.0003 μs for weighting and combination.

Keywords Wind speed prediction, Box & Jenkins methods, Artificial neural networks, Hybrid models, Ensembles, Genetic algorithm

Wind is a clean and inexhaustible source of energy. Research on wind energy systems has increased significantly. This growth is driven by the damage caused by fossil fuels[1,2] and their economic and environmental impact[3–5], with studies demonstrating the need for renewable alternatives across various applications including solar-powered systems[6]. According to the Energy Policy Tracker [http://www.energypolicytracker.org/] in 2021, Brazil invested $80.89 million in wind energy, behind only the United Kingdom, which invested $256.41 million.

Since wind speed forecasting is crucial to address the challenges associated with its unpredictable nature, different approaches are applied depending on the forecast horizon[4,7,8]. Wind speed forecasting can be classified into different categories according to the time horizon, including very short-term, short-term, middle-term, and long-term.

Very short-term forecasting covers the range of seconds to a few minutes. This forecasting type is particularly relevant in wind energy contexts. It provides crucial information for immediate regulatory actions and real-time turbine control. The system adjusts turbine settings based on real-time data from sensors such as anemometers and wind vanes.

Short-term forecasting covers minutes to a few hours and is essential for power grid planning and load management decisions and intelligent load-related decisions. Medium-term forecasts, which range from hours to a few days, are important for operational security in the electricity market, energy negotiations, and generation

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decisions. Long-term forecasts, which can span days to months, are necessary for reserve planning, maintenance scheduling, operational cost optimization, and operation management[9,10].

Numerous approaches have been developed to address the varying forecasting needs across different time horizons. These forecasting approaches can be classified into the main categories[11]: physical models[12], which will not be addressed in this research; statistical models[13]; machine learning (ML) models[14]; and hybrid models[15,16]. In addition, it is common to use model combination techniques, called ensembles, which integrate the forecasts generated by different predictive models[17,18].

Statistical models, such as the Autoregressive Integrated Moving Average (ARIMA)[19] and ARIMAX, use historical data patterns. Although they have been used for decades, these linear models are still used in wind speed forecasting[20–25]. However, assuming a linear relationship between the past data decreases their accuracy in time series with non-linear patterns[17].

Machine learning (ML) models have also been applied to wind speed forecasts[26,27], presenting relevant results in terms of precision. These models, such as Support Vector Regression (SVR)[28] and Artificial Neural Networks (ANNs), can handle complex patterns and non-linear relationships. Recent advances in machine intelligence for various applications, including renewable energy forecasting, have been comprehensively covered in research compilations such as[29], highlighting the growing importance of ML approaches across multiple domains.

Despite the advantages of ML models, their application to real-time series may not guarantee accurate results due to problems like overfitting (learning noise rather than patterns) and model misspecification (using wrong model type)[17]. Hybrid systems have been proposed to mitigate these problems and improve prediction accuracy. The combination of linear statistical models and ML applied at specific stages creates hybrid systems that aim to unite the strengths of these approaches[15,30]. The premise is that after forecasting a time series using a linear approach, the resulting residual (the difference between actual observed values and the linear model's predictions) consists primarily of non-linear components[31].

However, hybrid models may not always reduce modeling complexity. They can introduce several challenges. These include the need for multiple parameter adjustments and extensive error analysis. Additionally, they may require longer processing times and present significant challenges in parameter optimization[32–34].

Additionally, ensemble-based architectures combine predictions from different models for the same time series pattern[17,18]. By merging the outputs of different models, these methods improve overall accuracy. The key idea is that the models should be accurate yet diverse, so their strengths can balance out each other's weaknesses.

While it is essential to consider issues such as diversity (models using different approaches or capturing different patterns) and non-correlation between outputs (models making independent errors that don't systematically occur together) when using the ensemble technique, it is also necessary to answer some important questions: How should the outputs of the models be combined? Should all available predictive models be included? If not, which models should be chosen, and how should this selection be performed effectively?

It is worth noting that some researchers have started to incorporate classical optimization algorithms to perform such tasks, such as genetic algorithms (GAs). Works such as[35] used GA to select among several multilayer neural networks (MLP) which ones would be selected to be part of the ensemble to predict the price of Bitcoin. Similar to the previous method, the authors[36] used MLPs, followed by an ensemble method with weight adjustment to model the resonant frequency of rectangular micro-strip antennas (MSAs). Although the mentioned works applied ensembles with GA, it is clear that the base models used to form the ensemble belong to ML models. Therefore, there is an important trade-off between the use of linear models and ML models that extends to wind speed prediction.

In addition, it is possible to mention multi-objective optimization algorithms, which have become popular in dealing with complex problems involving multiple conflicting criteria. An example is the work of[7] who used a combined model that integrates several linear and non-linear prediction methods, employing a modified multi-objective optimization algorithm (MMODA) to determine the weights of the individual models. They provide a more detailed balance adapted to multiple performance demands. However, these algorithms face important scalability challenges. Parameter modifications can increase computational complexity. This makes MMODA less efficient in terms of runtime, especially for large-scale problems.

GA stands out as an efficient alternative for several reasons. It offers simplicity and flexibility in different optimization problems. The algorithm does not require excessive parameter adaptation. This characteristic enables operation with greater population diversity. Consequently, it favors optimal solutions and effective weight adaptation in complex scenarios[37,38].

Unlike approaches that may struggle to handle large search spaces or increase computational cost exponentially as problem complexity grows, GA adapts well to complex, multi-parameter problems and lends itself well to parallel implementation. This feature allows it to be applied in different contexts without losing efficiency, even in large-scale systems or when optimizing a large number of parameters, as occurs in complex ensembles.

Furthermore, GA has the advantage of not requiring mathematical derivatives, and is robust even on complex error surfaces, which is particularly relevant when working with diverse ensembles of predictive models. Thus, the use of a GA to assign weights to model output offers a balance, providing an optimized ensemble model adapted to the context of wind speed forecasting.

In this perspective, we used a GA in a previous work to combine linear models for WTI Oil Price Forecasting. Where different combination approaches were tested, including the simple mean, the median, the pseudo-inverse Moore-Penrose, and weighted combinations using a Genetic Algorithm (GA) and Particle Swarm Optimization (PSO)[39], and, given the success of the experiment, we propose the use of GA to optimize the weights in the ensemble in the context of wind speed forecasting.

The choice of a GA for optimizing the weights in the ensemble is justified by its ability to explore large solution spaces and avoid local minima[40,41], which is a significant advantage compared to methods such as the simple weighted average and the median, which assume a linear and fixed relationship between the models.

While GA-based ensemble optimization has been explored in forecasting contexts, our work presents three key novel contributions that distinguish it from previous approaches. First, we propose a systematic integration of heterogeneous model architectures–combining linear statistical models (AR, ARMA), multiple neural network types (MLP, RBF, ESN), and hybrid error-correction models–within a single ensemble framework. This approach maximizes model diversity beyond typical studies that combine models of the same family, improving forecast accuracy by accounting for the complex and unpredictable nature of wind data. Second, we apply this framework to minute-by-minute wind speed forecasting across five Brazilian cities with distinct climatic characteristics, demonstrating robustness and generalizability across geographically diverse locations. Third, we provide comprehensive computational efficiency analysis, showing that despite training complexity, the framework achieves real-time prediction capability (0.0003 µs for weight combination), making it practically viable for turbine control systems requiring immediate responses to address the unique challenges posed by the variability of wind conditions.

The proposed model, using these improved forecasts, can be integrated into the control systems of wind turbines. The energy processing unit will use the wind forecast to configure the control system's working point (rotation, pitch, torque, yaw, and captured power), with the aim of maximizing power in different wind conditions/speeds[42–44]. In this perspective, this article aims to focus on wind speed prediction to provide information for wind turbine control systems.

To verify the hypothesis of success in using the GA-based ensemble, this research aims to systematically compare the effectiveness of the proposed framework with various modeling approaches:

- *Linear models:* Autoregressive (AR) and Autoregressive with Moving Average (ARMA) models are selected as fundamental baseline models widely used in wind forecasting literature, providing essential linear trend capture and establishing performance benchmarks for comparison with more complex approaches[45];
- *Artificial Neural Networks:* Models such as Multilayer Perceptron (MLP), Radial Basis Function (RBF) and Echo State Networks (ESN) are selected for their complementary capabilities in modeling non-linear patterns: MLP for universal function approximation of complex wind dynamics, RBF for capturing localized non-linearities in rapid wind changes, and ESN for computational efficiency in real-time temporal dependency modeling[46];
- *Hybrid models:* Six combinations (AR+MLP, AR+RBF, AR+ESN, ARMA+MLP, ARMA+RBF, ARMA+ESN) are systematically included to test the error-correction hypothesis across different linear-nonlinear pairings, ensuring comprehensive evaluation of decomposition-based forecasting strategies where linear models capture trends and neural networks model residual non-linear components[47];
- *Ensembles:* Mean and median ensembles are included as non-trainable baseline methods to validate whether the proposed GA-based optimization provides meaningful advantages over simple combination strategies, while the complete set of linear, neural, and hybrid models ensures diverse input for ensemble learning[48].

In total, the study evaluates 14 forecasting models to improve the accuracy of wind speed forecasts. This selection ensures representation of major forecasting paradigms (linear, non-linear, hybrid, ensemble) while providing sufficient diversity to test the GA-based ensemble's ability to leverage complementary model strengths across different wind speed patterns and geographical locations.

These models are applied to five wind speed time series from five major Brazilian cities. Employing a one-step-ahead forecasting methodology using minute-by-minute data, the research highlights the critical importance of high-resolution temporal data.

This paper is organized as follows: Section Related Works reviews the relevant literature on the research topic. Section Theoretical background (Models) describes the fundamental principles of the selected models. Section Proposed Framework comprehensively describes the GA-based ensemble. Section Materials and Methods describes the methodology employed, while Section Case Study discusses case studies, focusing on the characteristics of the dataset employed. Section Results and Discussion presents the computational results accompanied by a critical analysis. Finally, Section Conclusion summarizes the main conclusions and points out directions for future research.

Related works

Wind speed forecasting has been a hot topic in the last decade due to the prominence that wind generation has gained. Linear models have been explored in wind speed forecasting applications[13]. studied ARIMA and WT-ARIMA models and proposed a new repeated WT-based ARIMA (RWT-ARIMA), which has improved accuracy for very short-term wind speed forecasting. They showed its superiority through a comparison with the benchmark persistence model[49]. used fractional-ARIMA models to predict wind speeds on long-term horizons. The results indicated significant accuracy improvements obtained with the proposed models compared to the persistence method.

The current literature has presented the importance of ML models to handle time series forecasting. The work of[50] proposed the use of a Novel Multiscale RBF (MSRBF) network to predict the Dst index, which determines the main characteristics of the geomagnetic field in magnetic storms. The investigation demonstrates that the model is more flexible for describing complex non-linear dynamical systems.

Beyond wind forecasting, ML techniques have also proven effective in broader meteorological applications, with comparative studies showing that ensemble methods like XGBoost and CatBoost achieve superior performance in weather prediction tasks[51], demonstrating the versatility of ML approaches across different atmospheric phenomena.

The application of explainable AI techniques has gained attention in renewable energy forecasting, with[52] demonstrating the use of machine learning models like decision trees, random forest, and extreme gradient boosting for global horizontal irradiance prediction while providing interpretability through explainable AI frameworks.

Similarly, neural networks have also been applied to wind speed prediction[53]. performed a comparative study among Multiple Linear Regression, LASSO Regression, Support Vector Regression (SVR) and Multilayer Perceptron (MLP) for a database from the USA. The SVR performed better in this case study[27]. proposed an approach to predict the wind speed for locations lacking historical data. Instead, they trained the models using the data supplied by 55 wind measuring stations and tested their prediction ability.

[54] proposed Long Short-Term Memory (LSTM) neural networks to improve the accuracy of forecasting models for short-term wind speed. Their results demonstrated better efficiency than other neural approaches.

[55] conducted a comparative analysis between Bi-LSTM and Uni-LSTM algorithms for wind speed estimation in Rajasthan, India, demonstrating that GRU models outperformed both LSTM variants in terms of R^2 accuracy and error metrics.

More sophisticated single models, such as the Boosting Wavelet Neural Network, have also been developed to address the same task[56]. The ANN is trained using bioinspired metaheuristics (genetic algorithm [GA] and particle swarm optimization [PSO]) and coordinate dictionary search optimization. The same research group proposed the use of a hybrid deep learning approach that combines a time series decomposition algorithm with a gated recurrent unit (GRU)[57]. The decomposition was performed using the complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) and (2) wavelet packet decomposition (WPD).

Similarly[58], demonstrated the effectiveness of hybrid deep learning architectures by combining CNN, LSTM, and GRU for global horizontal irradiance estimation, achieving superior performance compared to standalone models across multiple geographical locations in India.

Ensemble methods that combine more than one model response are also used for wind speed forecast[7]. proposed a three-stage approach, where the final stage applies reinforcement learning by integrating three types of deep neural networks. They concluded that the ensembles are effective and work better than the traditional optimization-based ensemble method. In all cases, it achieved accurate results, outperforming state-of-the-art models in accuracy[59]. proposed a new prediction method that combines the wavelet transform, the echo state network, and the ensemble technique. The ensemble component mitigates model misspecification and data noise, with simulation results demonstrating the superiority of the proposed method over benchmark algorithms.

More recently[60], proposed an interpretable stacked ensemble model using ridge regressor with multiple base learners (CatBoost, Gradient Boost, MLP, and Random Forest) for wind speed estimation, demonstrating significant error reductions of 19.89% in RMSE and 24% in MAE while providing model interpretability through SHAP analysis.

[61] proposed a hybrid model for wind speed prediction using a linear model - ARIMA - and machine learning approaches – ELM, SVM, and LSSVM (Least Square SVM), which are combined using a Gaussian process regression combiner. The data were pre-processed using the empirical wavelet transform (EWT). The same research group proposed a model that uses the CEEMDAN denoising strategy to decompose wind speed data, removing noise and reconstructing a cleaner series. Four neural networks are selected, BPNN, GRNN, RBF, and ELM, to process these denoised data after extensive parameter adjustments. The Multiobjective Gray Wolf Optimizer (MOGWO) algorithm is then used to assign optimized weight coefficients, combining neural networks into a weighted model[62].

Hybrid systems based on error correction that combine linear and non-linear models have been used. This model combines the forecasts of the linear (time series) and non-linear (residual series) models. The significant difference lies in choosing the linear, non-linear, and combination models to be used. For example[63], used an ARIMA + MLP and an ARIMA + Kalman Filter for multistep prediction. To predict wind speed with long-term data[64], introduced a SARIMA and Least Squares Support Vector Machine (LSSVM, SARIMA + LSSVM)[16]. Alternatively, relevant results were obtained from hybrid systems that use the non-linear combinations[65]. This strategy was only recently used in wind speed forecasting scenarios. In[30], an SVR was used to find the best combination of linear and non-linear forecasts. Furthermore[15], used a Long Short-Term Memory (LSTM) network to model residuals, as it can capture long-range dependencies in the series.

Finally, it is important to mention surveys and reviews that have presented the effectiveness and robustness of machine learning and neural networks in real-world problems[66]. presented an interesting review on the use of ANNs in pattern recognition, showing the effectiveness of these models in the current literature[67]. presents the importance of ANNs in fields like computing, science, engineering, medicine, environmental, agriculture, mining, technology, climate, business, arts, etc. They also indicate as a future perspective the use of combination models. The same research group also published a paper that describes the importance of correctly defining the inputs of a system, the feature selection task[68]. The authors explain in detail the approaches and trends in the field, as did[69] in their investigation focused on bioinspired methods. Tables 1 and 2 presents some of the many works in the area, selected to provide a broad yet representative overview of the diverse methodologies applied in wind speed forecasting. These works were chosen to highlight the main advances, comparative studies and combinations of methods, promoting our proposal.

Materials and methods

This section presents the comprehensive methodological framework employed in this study, covering data acquisition, pre-processing techniques, model development, and evaluation procedures.

Authors	Publication Year	Models	Contribution	Limitation
[49]	2009	f-ARIMA; ARIMA	The differentiation parameter *d* takes fractional values	The paper does not address potential challenges in model selection and parameter estimation that may arise in practice, especially in real-time forecasting scenarios.
[64]	2011	LSSVM; Seasonal Auto-Regression Integrated Moving Average (SARIMA); ARIMA	The paper presents a new hybrid model that combines the algorithms (SARIMA) and (LSSVM)	Traditional models like SARIMA, which assume a linear form, may not adequately capture the non-linear patterns present in wind speed time series.
[63]	2012	ARIMA-ANN and ARIMA-Kalman	The paper presents two new hybrid models, ARIMA-ANN and ARIMA-Kalman	Complexity in the modeling process. This complexity may require more computational resources and expertise for implementation, limiting practical applications in some environments.
[61]	2015	Empirical Wavelet Transform (EWT); Gaussian Process Regression (GPR); ARIMA; ELM; SVM; LSSVM (Least Square SVM)	The hybrid model provides probabilistic forecasts	The integration of various forecasting models using GPR may introduce complexity in terms of model management and computational requirements.
[27]	2018	MLP	Approach to determine wind speed in locations without previous data.	The paper does not explore the scalability and practical feasibility of these models in resource-limited environments.
[16]	2018	Holt-Winters; ANN; ARIMAX;	Development of hybrid models and inclusion of exogenous variables such as temperature, pressure, and precipitation.	The authors do not explicitly mention any limitations related to the computational resources used in the study.
[13]	2019	RWT-ARIMA; ARIMA	New forecasting model, the ARIMA model based on repeated wave transformation (RWT-ARIMA)	Involves multiple levels of decomposition and modeling, which can increase complexity and computational times.
[56]	2019	Wavelet Neural Networks; GA; PSO; Coordinate Dictionary Search Optimization (CDSO); Sic Function.	Structure for modeling high-dimensional data; Training with evolutionary and derivative-free algorithms	Alternative wave functions; there may be better alternative options, such as Gaussian waves or radial basis functions; lack of comparison with other methods.
[59]	2019	ESN; ensemble average	The paper presents a new wind power forecasting method that combines wavelet transformation, echo state network (ESN) and ensemble	The model struggles when the approximate subsignal changes significantly. This indicates a limitation in the model's ability to handle rapid fluctuations in wind power data, suggesting a need for greater refinement in predicting these variations.

Table 1. Summary of the presented studies.

Authors	Publication Year	Models	Contribution	Limitation
[54]	2020	LSTM	Hybrid model Crow Search Algorithm (CSA), Wavelet Transform (WT), Feature Selection (FS) based on entropy and mutual information (MI), and LSTM	Complexity in implementation and understanding. The integration of several components may also lead to increased computational requirements and longer processing times.
[30]	2020	ARIMA; ARIMAX; MLP; SVR; RBF	The paper presents a new hybrid system that combines linear statistical models with machine learning models	Although the paper proposes a non-linear combination of linear and non-linear models, it highlights that the most suitable function to combine these forecasts is still an open question.
[15]	2021	ARIMA, ARIMAX, SARIMA, SARIMAX; LSTM	The main contribution is the creation of a flexible hybrid system that combines linear and non-linear models to improve wind speed forecasting accuracy; The paper presents an evolutionary search process to identify the ideal combination of linear and non-linear forecasts, including the use of exogenous variables.	The use of LSTM for residual modeling, while beneficial for capturing non-linear patterns, results in greater computational complexity compared to other models like MLP and SVR.
[57]	2021	Gated Recurrent Units (GRU); LSTM	Integration of EMD Time Series Decomposition and GRU	EMD may lead to inaccurate forecasts because the frequency and amplitude of the first intrinsic mode function (IMF) can fluctuate significantly, affecting the accuracy of the forecast.
[53]	2021	Multiple Linear Regression; SVR; MLP	The use of LASSO regression shows an attempt to mitigate overfitting and improve model generalization	There is a lack of discussion about the scalability and practical feasibility of these models in resource-limited environments.
[7]	2021	ARIMA; Back Propagation Neural Network (BPNN); Ensemble Neural Network (ENN) with weights; Extreme Learning Machine (ELM); General Regression Neural Network (GRNN)	Structure for modeling high-dimensional data; training with evolutionary and derivative-free algorithms	The multiobjective optimization algorithms used have inherent limitations, including a tendency to fall into local optimization and slower convergence speeds in the later stages of the algorithm.

Table 2. Summary of the presented studies (Cont.).

Beginning with an overview of the general methodology, we present the specific case study that provides the real-world context for our research.

The characteristics of the dataset, analysis procedures, and pre-processing steps are detailed, followed by the theoretical foundations of the linear and neural network models used in this work. We then describe our proposed prediction framework, along with postprocessing analysis techniques.

Methodology

This study on an extensive comparison of time series forecasting models can be described in 8 main steps, as follows Fig. 1 illustrates.

The workflow begins by analyzing the raw data from each of the five datasets individually to identify key patterns and characteristics. These datasets will be processed and evaluated one at a time, allowing for a detailed comparative analysis later.

This includes performing the Augmented Dickey-Fuller (ADF)[70] test to check for stationarity, which is a key requirement for many time series forecasting models. Additionally, analysis may involve checking for missing data and outliers so that the pre-processing steps required for modeling can be done.

If the data is stationary and there is no need to deal with missing data or outliers, no pre-processing is required. The data is then split. On the other hand, if the data is non-stationary, it is first deseasonalized before splitting.

Each dataset was split so that the last 20% of the samples were used to test, the first 60% for training, and the other 20% for validation. The validation set served multiple critical purposes in our modeling framework: (1) for linear models (AR and ARMA), it was used to select optimal model orders through grid search evaluation; and (2) for neural networks (MLP, RBF, ESN), it guided hyperparameter tuning and prevent overfitting. With this data partitioning in place, we then proceeded to identify the input for predictions using both linear and non-linear models.

The processed data then feeds into the Model Application phase, where forecast models are applied.

Regarding input selections, the order of the linear models (AR(p) and ARMA(p, q)) was determined considering a grid search from 1 to 11. To identify the most relevant past observations (lags) for individual neural models, the Partial Autocorrelation Function (PACF) is employed on the training set. This allows the selection of informative lags as inputs for the prediction models[68].

As for model fitting, more specifically with respect to the input data of the linear models, the following configuration was considered: AR(9) and ARMA(2,2). Interestingly, the best configuration was the same for all datasets.

After these are executed, post-processing techniques refine the outputs before generating the final forecast data.

The framework incorporates two parallel paths: (1) forecast data is directly benchmarked against ground truth values to assess individual model performance, while (2) ensemble methods combine multiple model outputs to improve forecast robustness and then subject them to evaluation.

To carry out this study, a computer equipped with a 3.8 GHz AMD Ryzen 7 5700G processor, with 8 cores and 16 logical processors, in addition to 32 GB of installed RAM was used. The implementation of the forecasting models was done in the Python language, taking advantage of its advanced libraries such as statsmodels, sklearn, and keras.

Case study

To compare the forecasting models, five time series related to wind speed from important cities in Brazil are selected. The data originate from anemometers 10 meters from the ground located in Brasília (S1), Florianópolis (S2), Natal (S3), Petrolina (S4), and São Luís (S5). These stations are partners of the SONDA project (National Environmental Data Organization System) operated by the Brazilian agency called National Institute for Space Research (INPE)[http://sonda.ccst.inpe.br/]. All data series encompass one-minute interval measurements collected from January 1 st to January 31 st, 2019 (all data in meters per second, m/s).

The series of 5 cities with different geographical characteristics were used, which can be seen in Table 3, in which annual wind speed values are based on the Brazilian Center for Wind Energy, 1998.

S1 and S2 are coastal towns, S5 is a riverside town. The other two are cities further inland, with S4 in a flat region 700 km from the coast and S1 on the central Brazilian plateau.

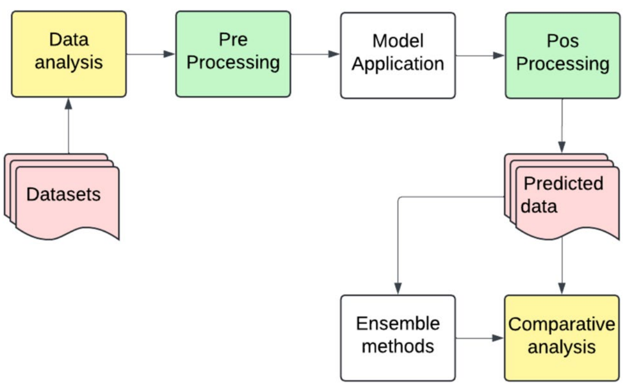

Fig. 1. Flowchart of the Modelling Process.

City	Series	Latitude	Longitude	Elevation [m]	Annual average [m/s]
Brasília	S1	−15.7213681	−48.0197587	1172	5 - 6
Florianópolis	S2	−27.5705976	−48.799918	3	< 5
Natal	S3	−5.801411	−35.3046079	3	< 5
Petrolina	S4	−9.3731899	−40.5879725	10	6 - 7
São Luís	S5	−2.5604581	−44.340523	3	> 8.5

Table 3. Geographical indication of the cities where the series were collected and climate information.

Series	Mean [m/s]	STD [m/s]	Min. [m/s]	Max. [m/s]	CV
S1	1.0271	0.7146	0.0	5.10	0.69
S2	2.1154	1.6810	0.0	11.63	0.79
S3	5.9776	2.6976	1.2740	87.40	0.45
S4	2.9464	1.3455	0.0	13.55	0.45
S5	2.2264	1.4056	0.022	9.32	0.63

Table 4. Descriptive statistics of the series wind speed addressed.

Dataset	ANNs	AR+ANNs	ARMA+ANNs
S1	[1, 2, 3, 4, 5, 6, 7, 8, 9, 12]	[5, 6, 7, 8, 11, 16, 17, 22, 24, 30]	[2, 3, 5, 7, 10, 16, 17, 22, 27, 30]
S2	[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]	[5, 6, 7, 8, 9, 10, 11, 24]	[3, 9, 10, 11, 14, 19, 21, 22, 24, 27]
S3	[1, 2, 3, 4, 5, 6, 7, 13, 14, 15]	[8, 9, 14, 15, 16, 21, 22, 23, 24, 30]	[7, 8, 14, 15, 16, 21, 22, 23, 24, 30]
S4	[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]	[3, 4, 5, 12, 15, 17, 21, 23, 27, 29]	[9, 17]
S5	[1, 2, 3, 4, 5, 6, 7, 8, 10, 12]	[5, 6, 7, 10, 12, 15, 16, 18, 20, 30]	[3, 9, 11, 12, 15, 16, 18, 20, 28]

Table 5. Lags selected by Partial Autocorrelation Function (PACF).

Dataset, data analysis and pre-processing

Each of the datasets employed in this study consists of 44,640 samples, with the exception of Series 1 (S1), which contains 270 missing samples. In this instance, the decision was made to suppress the gaps in the series by excluding the missing data points rather than applying more sophisticated imputation techniques.

This approach was chosen to maintain the primary focus of the research on the comparative analysis of the selected forecasting models, without introducing additional complexities associated with missing data handling strategies. Given that the missing data represents only a small fraction–approximately 0.6%–of the total dataset, the impact of this exclusion on the overall analysis is expected to be minimal.

Therefore, the number of samples for each set is 26784 for training, 8928 for validation, and 8928 for testing. Considering S1, these values are respectively 26622, 8874, and 8874. Table 4 presents the descriptive statistics of the each dataset, in which STD is the standard deviation and CV means the coefficient of variation.

The cities corresponding to series S4, S3, and S5 are located in the Northeast region of Brazil, with the last two series referring to coastal cities. They are close to the Equator line and present the highest wind speed average throughout the year. S3, indeed, is the more eastern city of the county, in a way that its average wind speed is the highest. S1 is the country's capital and is located in the center of Brazil. S2 is another coastal town located in the Brazilian south. Its wind speed average is about double that of S1. These cities were chosen according to the variability between the different climate zones in Brazil with a view to studying the expansion of Brazilian renewable energy (wind).

To model wind speed patterns across these geographically distinct locations, the time series structure of each dataset was analyzed. The Partial Autocorrelation Function (PACF) was employed to identify relevant temporal dependencies, ensuring optimal input selection for the forecasting models.

Table 5 presents the lags selected by the Partial Autocorrelation Function (PACF) for each dataset and forecasting approach. These lags were used as inputs for ANNs and hybrid models.

The first column, where the delays of the real data are provided as inputs to neural networks (ANNs), shows the selection of lags 1 through 9 across various datasets, suggesting that the neural networks are primarily capturing short-term dependencies in the wind series. These short-term lags likely correspond to rapid fluctuations in wind speed, which are characteristic of the volatile and intermittent nature of wind, particularly in inland locations.

On the other hand, the AR+ANN and ARMA+ANN columns, which demonstrate the selected delay on the forecast residue of the linear models, show the inclusion of longer lags, such as lag 30. Which may indicate a periodic non-linear pattern.

This pattern suggests that while the AR and ARMA models were effective in capturing the overall linear structure, the ANN component was necessary to model the remaining complex interactions, particularly those related to local atmospheric conditions.

Following the identification of the relevant lags for the non-linear models, the data prepared for the neural networks is scaled to the range [−1, 1]. This is mandatory since we are using the hyperbolic tangent as an activation function[71].

Theoretical background (models)

In this study, a range of linear models (AR and ARMA) and four distinct neural network architectures were employed to model and predict time series data relevant to wind energy forecasting. The selection of these methods is particularly motivated by their ability to address the unique challenges present in the wind energy domain, where both linear and non-linear patterns are observed due to the stochastic nature of wind.

The AR and ARMA models were selected for their established efficacy in capturing linear dependencies in time series data. These models are widely recognized in the wind energy community as reliable tools for short-term forecasting, providing a solid baseline for direct comparisons between traditional linear methods and more complex non-linear approaches.

The use of MLP and RBF networks is motivated by their capacity to model non-linear relationships[71], which are frequently encountered in wind speed and power data due to the intricate interactions between atmospheric variables. MLP is valued for its flexibility and its capacity to approximate a wide range of functions, making it suitable for capturing the broader, non-linear trends in wind energy data. RBF networks, on the other hand, are particularly effective in capturing localized nonlinearities, which are crucial for accurately predicting wind patterns that exhibit rapid changes over time.

The selection of ESN is driven by their demonstrated ability to achieve robust predictive performance with reduced computational overhead, a significant consideration in the wind energy sector where real-time forecasting is often required. Additionally, ESN's dynamic reservoir of recurrent connections allows it to effectively model the temporal dependencies that are characteristic of wind energy time series, where past wind conditions can significantly influence future behavior.

Furthermore, the development of ensemble and hybrid models based on error correction strategies is particularly relevant for the wind energy community, as these approaches are designed to enhance predictive accuracy by integrating the strengths of individual models. Wind energy forecasting often benefits from such strategies, as they allow for the mitigation of individual model weaknesses, leading to a more reliable prediction of wind patterns and, consequently, more efficient energy production and grid management.

Below, detailed descriptions of the methods used in this work are presented. Considering: a) single models: Autoregressive (AR) and Autoregressive and Moving Average (ARMA) models, Radial Basis Function (RBF) network, Multilayer Perceptron (MLP), Echo State Networks (ESN); b) Heuristic algorithms: Genetic Algorithm (GA); c) combination approaches: ensembles and hybrid models.

Autoregressive models

The Autoregressive (AR) model is formally defined as a linear combination of past observations, where each observation Z_t is expressed as a weighted sum of its previous *p* values, or lags, resulting in an AR(*p*) process[19]. In this model, the parameters $\phi_1, \ldots, \phi_p$ represent the weights assigned to each lagged observation, while ε_t denotes a white noise error term that captures random fluctuations.

Determining the optimal values of the parameters ϕ_i involves solving a set of linear equations derived from the model's autocorrelation function. This system of equations is commonly known as the Yule-Walker equations, which provide a closed-form solution for estimating the parameters of the AR model. The Yule-Walker method leverages the stationarity assumption of the time series, ensuring that the statistical properties, particularly the mean and variance, remain constant over time, thereby simplifying the estimation of the model coefficients.

Autoregressive and moving average models

The Autoregressive Moving Average (ARMA) model synthesizes the Autoregressive (AR) and Moving Average (MA) models, offering a flexible approach to time series modeling by combining their strengths. Although the AR model relies solely on past observations (lags), the MA model incorporates past forecast errors (white noise terms). The ARMA model thus captures linear relationships between the current value, past values (AR component) and past errors (MA component).

Unlike the AR model - where parameters can be estimated analytically using the Yule-Walker equations - the inclusion of the MA component in the ARMA model complicates parameter estimation. This necessitates iterative optimization methods, such as Maximum Likelihood Estimation (MLE), which identifies parameters that maximize the likelihood of observing the given data. MLE is favored for its asymptotic properties: as the sample size increases, the estimators become consistent, efficient, and normally distributed[19].

Multilayer perceptron

The Multilayer Perceptron (MLP) is a class of feedforward artificial neural networks (ANNs) characterized by a layered architecture composed of an input layer, one or more hidden layers, and an output layer. Each layer consists of a set of artificial neurons, or nodes, where each neuron in a layer is fully connected to all neurons in the subsequent layer. The MLP is particularly renowned for its ability to model complex, non-linear relationships due to its hierarchical structure and the use of non-linear activation functions. This capability makes the MLP a universal function approximator, applicable to a wide array of tasks such as non-linear mapping, pattern recognition, process identification and control, time series forecasting, and system optimization[72].

Training an MLP involves the optimization of its synaptic weights to minimize a predefined loss function, typically the Mean Squared Error (MSE) in regression tasks or cross-entropy in classification tasks. The training process requires iterative adjustment of these weights to allow the network to learn from the data. This adjustment is achieved using the Stochastic Gradient Descent (SGD) algorithm, which minimizes the loss function by updating the weights in the direction of the negative gradient. Calculating this gradient is performed using the backpropagation algorithm, a fundamental technique in neural networks that computes the loss function gradient with respect to each weight, applying the chain rule of calculus through the layers of the network[73].

The backpropagation algorithm operates in two phases: forward pass and backward pass. During the forward pass, the input data is propagated through the network layer by layer, generating a predicted output at the output layer. The network output is then compared to the target output, and the resulting error is calculated. In the backward pass, this error is propagated back through the network, layer by layer, in reverse order. The gradient of the loss function with respect to each weight is computed, and the weights are updated accordingly using the gradient descent optimization rule. This iterative process continues until the network converges, that is, when further iterations result in negligible changes in the loss function, indicating that the MLP has learned to approximate the underlying function that maps inputs to outputs.

Radial basis function network

Radial Basis Function (RBF) networks represent a distinct class of artificial neural network (ANN) architectures that differ significantly from the more commonly known multilayer perceptrons (MLP). Unlike MLPs, the RBF networks typically consist of only two computational layers, excluding the input layer: a hidden layer and an output layer. The hidden layer in an RBF network is characterized by its use of radial basis functions as activation functions, which are centered around specific points in the input space. The most widely used radial basis function in these networks is the Gaussian function.

The training process of an RBF network is typically conducted in two distinct phases. In the first phase, the parameters of the hidden layer are determined, specifically the centers c_i and the spreads σ_i of the radial basis functions. This phase is often performed using unsupervised learning techniques, such as clustering of K-means, which adjust the centers to optimize the coverage of the input space, thus ensuring that the radial basis functions are well positioned to capture the underlying structure of the data[74]. The variance σ_i is usually set based on the distribution of the input data within the clusters.

In the second phase, the output layer weights are optimized in a supervised learning process. This involves minimizing a loss function, such as the Mean Squared Error (MSE), by adjusting the weights that connect the hidden layer to the output layer. The linear combination of the Radial Basis functions, weighted by these optimized output weights, forms the final output of the network. This dual-phase training process allows RBF networks to effectively model complex, non-linear mappings while maintaining a relatively simple and interpretable structure.

Echo state networks

Echo State Networks (ESNs) are a specialized form of Recurrent Neural Networks (RNNs) that were introduced to address some of the challenges associated with traditional RNNs, particularly the difficulties in training due to vanishing and exploding gradients. The defining feature of ESNs is the presence of feedback loops within the hidden layer, referred to as the dynamic reservoir, which allows the network to retain and process information over time[75]. This structure is particularly advantageous for modeling sequences with temporal dependencies, such as those found in time series forecasting, where the temporal context is crucial for accurate predictions. In the original formulation of ESNs, these feedback loops are confined to the reservoir, distinguishing them from other RNN architectures where feedback might also be present in other layers.

The architecture of an ESN comprises three main components: the input layer, the dynamic reservoir (hidden layer), and the output layer. The dynamic reservoir consists of a large number of sparsely and randomly connected neurons that introduce non-linearity into the model. This reservoir acts as a complex, high-dimensional space where the input signals are dynamically transformed, allowing the network to capture intricate temporal patterns. The fixed nature of the reservoir weights during training is a key characteristic of ESNs; rather than adjusting these weights, the focus is on optimizing the output layer. The output layer receives signals from the dynamic reservoir and linearly combines them to produce the final output.

To optimize the output layer weights, ESNs utilize the Moore-Penrose pseudoinverse, a method that provides a stable and efficient solution for the linear regression problem posed by the output layer. This approach ensures that the output layer is finely tuned to map the complex representations generated by the dynamic reservoir to the desired output, making ESNs particularly effective for tasks requiring the modeling of temporal dependencies. The use of a fixed reservoir and the application of the pseudo-inverse for training contribute to the computational efficiency and stability of ESNs, enabling them to be applied effectively in real-time and large-scale time series forecasting applications.

Genetic algorithms

Genetic Algorithms (GA) are a class of optimization techniques inspired by the mechanisms of natural evolution, such as selection, crossover, and mutation. They are designed to efficiently explore complex solution spaces, generating diverse candidate solutions through evolutionary strategies such as selection, crossover, and mutation[76].

A GA begins with a randomly generated population of candidate solutions, each represented by a chromosome-like structure. These candidates are evaluated using a fitness function that quantifies their performance in relation to the problem at hand. The fittest individuals are selected to produce offspring through crossover, which combines elements of two parent solutions, and mutation, which introduces random variations to explore new areas of the solution space[77].

The iterative process of selection, reproduction, and mutation drives the population towards increasingly better solutions over successive generations. This evolutionary approach allows Genetic Algorithms to effectively navigate large, complex solution spaces and avoid local optima, making them particularly useful in scenarios where traditional optimization methods may struggle[78].

In summary, Genetic Algorithms leverage principles of natural evolution to optimize complex systems by iteratively refining a population of solutions. Their ability to balance exploration and exploitation in the search space makes them a powerful tool for solving a wide range of challenging optimization problems.

Hybrid models based on error correction

Hybrid models based on error correction have gained significant attention in recent years. Initially proposed by Zhang[17], this approach combines linear statistical methods with machine learning (ML) techniques to improve the overall performance of forecasts. The fundamental premise involves using a linear model for the primary time series prediction, while employing an ML model to capture and predict non-linear patterns present in the error series. The final forecast $\hat{Z}_t$ is obtained by summing the linear forecast component $\hat{L}_t$ and the non-linear residual prediction $\hat{N}_t$.

This dual-stage modeling approach offers two key advantages: it enhances the model's adaptability to new or unforeseen data patterns and significantly improves prediction accuracy through targeted non-linear error correction. By decomposing the forecast problem into linear and non-linear components, the hybrid model can effectively minimize the general forecast error[17].

In addition to Zhang's initial proposal, other hybrid models began to be used, using other machine learning model architectures and in this case, discarding the use of linear models, as in[79].

Ensembles

Ensemble methods represent sophisticated strategies that aggregate the results of multiple predictive models to improve overall system performance[80,81]. These methodologies capitalize on the inherent diversity among different models, which often exhibit distinct behaviors when exposed to identical inputs. The final output of the ensemble is typically generated through combination techniques such as averaging, voting, or the integration of an additional neural network, which produces a more robust and precise prediction[82].

Ensembles can be categorized into trainable and non-trainable types. In a non-trainable ensemble, the combination methods, such as averaging or voting, do not involve any additional learning process. The models' predictions are combined using predefined rules without further adjustments. In contrast, a trainable ensemble, such as those using stacking or neural network integration, involves an additional learning step. This step optimizes how the models' outputs are combined, typically through a meta-learner or an additional neural network that is trained to minimize prediction errors.

To achieve the full potential of an ensemble, it is essential that the individual models are diverse, as the primary goal of an ensemble is to improve upon already strong predictive results. This dual requirement ensures that the ensemble is not just a redundant combination but a synergistic enhancement, producing outcomes superior to those of any single model.

Proposed framework

This study proposes a GA-based ensemble of predictive models that aims to optimize the weighting coefficients for each model in the ensemble through a systematic, iterative process based on evolutionary computation. The final output is calculated as the weighted average of the individual model predictions, as represented by equation 1.

$$\hat{y} = \sum_{i=1}^{N} w_i \cdot \hat{y}_i, \tag{1}$$

In equation 1, $\hat{y}$ denotes the final prediction of the proposed framework, w_i represents the weights assigned to each model, and $\hat{y}_i$ refers to the individual predictions of the models. These individual predictions are structured into a matrix P 2, where each row of the matrix corresponds to a specific time point, and each column represents the predictions from a particular model.

$$P = \begin{bmatrix} \hat{y}_1(1) & \hat{y}_2(1) & \dots & \hat{y}_N(1) \\ \hat{y}_1(2) & \hat{y}_2(2) & \dots & \hat{y}_N(2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{y}_1(t) & \hat{y}_2(t) & \dots & \hat{y}_N(t) \end{bmatrix} \tag{2}$$

Therefore, the final prediction of the framework for all time points t can be expressed as a matrix multiplication between the prediction matrix P and the weight vector w, as given by Equation 3.

$$\hat{\mathrm{y}} = P \cdot \mathrm{w} \tag{3}$$

Here, $\hat{\mathrm{y}}$ is the vector that contains all the final predictions of the framework over time, P is the matrix of individual predictions, and $\mathrm{w} = [w_1, w_2, \dots, w_N]$ is the optimized weight vector.

The weights w_i are optimized using GA, with the objective of minimizing the prediction error of the ensemble[39,83]. In the context of this work, each weight w_i is associated with a specific predictive model, representing the relative importance of its predictions in the construction of the final proposed framework. The set of weights W is encoded in a chromosome, with the constraint that the sum of the weights is equal to one and that all the weights are non-negative. To optimize these weights, we employ single point crossover, mutation with a rate of 10%, and tournament selection, as detailed in Table 6.

The GA parameters were selected based on established best practices and preliminary experimentation. A population size of 20 individuals provides adequate genetic diversity while maintaining computational efficiency for the 14-dimensional weight optimization problem. The 50 generations typically ensure convergence to stable solutions, while the 10% mutation rate balances exploration and exploitation. Multiple independent runs (30) account for the stochastic nature of GAs and ensure statistical robustness of results.

The initial population consists of weighting sets, where each set represents a potential solution for the GA-based ensemble. The weights are initialized randomly and then normalized to ensure that their sum equals 1, as described by Equation 4.

$$\mathrm{w} = [w_1, w_2, \ldots, w_N], \quad \text{the} \quad \sum_{i=1}^{N} w_i = 1 \tag{4}$$

The fitness function evaluates the quality of each set of weights. The fitness of each individual is defined as the inverse of the MSE between the predictions of the proposed framework and the actual values, as expressed in Equation 5.

$$\text{Fitness}(\mathrm{w}) = \frac{1}{\text{MSE}(\mathrm{w})} \tag{5}$$

the $\text{MSE}(\mathrm{w})$ represents the Mean Squared Error (MSE) calculated between the predictions and the actual values, y. This process is iterated over multiple generations until the set of weights that minimizes the prediction error is identified.

The complete GA optimization process for the ensemble framework is illustrated in Figure 2, which provides a detailed schematic of the evolutionary algorithm workflow.

Given the iterative nature and computational steps of the proposed framework, it becomes imperative to evaluate the computational complexity of this optimization task.

The computational complexity of this optimization task is mainly dependent on the number of models indicated by n. With constant parameters for the number of generations, population size, and execution runs, the computational complexity is $O(n)$, indicating that the execution time of the GA-based algorithm scales linearly with the number of models in the ensemble.

For relatively small values of n (e.g., fewer than 1000 models), both runtime and memory consumption remain within manageable limits on standard computational resources.

The GA-based ensemble framework embodies the fundamental principles of the ensemble by leveraging the diversity of the model to improve predictive accuracy. By utilizing a Genetic Algorithm for weight optimization, this approach has relatively low computational overhead compared to more resource-intensive optimization techniques.

The novelty of this framework lies not only in the optimization mechanism but primarily in the strategic diversity of base models. Unlike previous ensemble approaches that typically combine models from the same family (e.g., multiple neural networks with different architectures or hyperparameters), our framework deliberately integrates fundamentally different modeling paradigms: classical time series models that capture linear autocorrelation patterns, neural networks that model complex nonlinear relationships, and hybrid systems that perform targeted error correction. This heterogeneous combination allows the GA to leverage complementary strengths–linear models provide stable baseline predictions, neural networks capture sudden fluctuations, and hybrids address systematic residual patterns–resulting in a more robust ensemble than homogeneous approaches could achieve.

Pos-processing and results analysis

With training and validation completed, data prediction is performed according to the test set. So the outputs are subjected to inverse scaling. This process converts the scaled outputs back to their original scale.

Following inverse scaling, any treatments applied to make the series stationary can be undone. This ensures that the final output reflects the true characteristics of the original data series, enabling a direct comparison between the predicted and actual values. By reversing any deseasonalization or other pre-processing steps, the integrity and interpretability of the predicted series are maintained.

The data is now ready to evaluate the performance of the prediction models, in which the metrics are used: Mean Squared Error (MSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2). These

Parameter	Description	Value
`pop_size`	Population Size	20 individuals
`num_generations`	Number of Generations	50
`mutation_rate`	Mutation Rate	10%
`num_executions`	Number of Independent Runs	30 runs

Table 6. Genetic Algorithm Parameters.

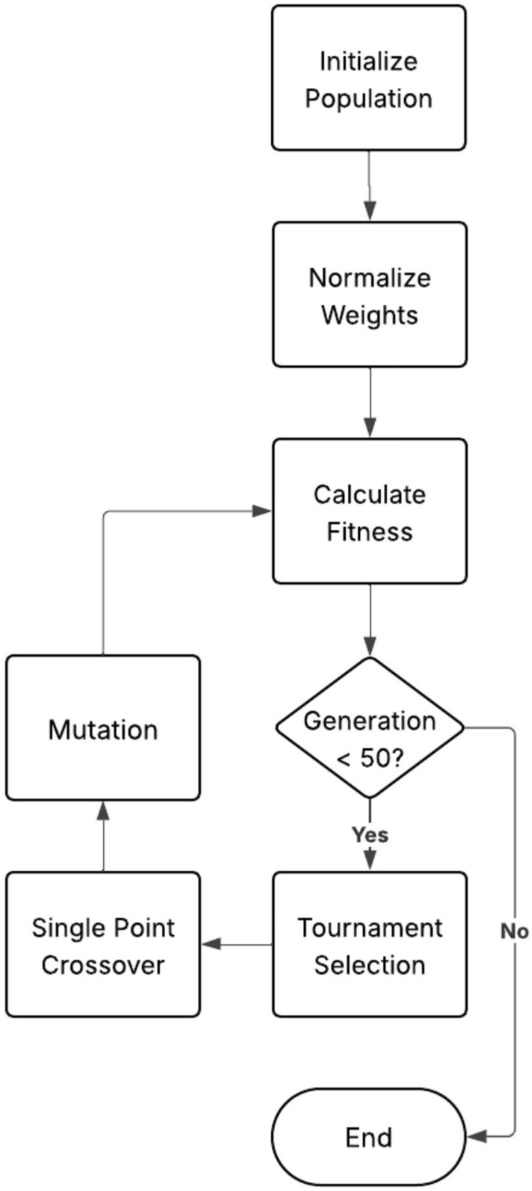

Fig. 2. Schematic diagram of the GA optimization process for ensemble weight determination.

metrics, computed using scikit-learn's metrics module, provide a comprehensive and balanced assessment of the model's performance. MSE emphasizes the importance of penalizing larger errors by squaring the differences between predicted and actual values, making it particularly useful when outliers or significant deviations are critical to address. In contrast, MAE offers a straightforward interpretation of the average error, treating all errors equally and providing a clear picture of the model's overall accuracy without being overly influenced by outliers. Together, MSE and MAE ensure that both the typical error and the impact of larger errors are adequately captured.

Finally, R^2 complements these metrics by evaluating the proportion of variance in the dependent variable that is explained by the model, offering a clear indication of the model's overall explanatory power. This combination of metrics ensures a robust evaluation, covering error magnitude, bias, and the model's ability to explain variability, making it a well-rounded approach for assessing predictive performance.

To facilitate the evaluation of the models' performance, a table of comparative percentages is included that shows how much greater the error of each model is compared to the model with the best performance in each case, represented by.

Considering the errors generated by the models, the Friedman test was applied to evaluate whether the performance of the winning model was statistically significantly different from the other models[84].

Model	S1	S2	S3	S4	S5
AR	0.0823	0.2861	0.9433	0.5073	0.2417
ARMA	0.0812	0.2830	0.9243	0.4973	0.2399
ESN	0.0810	0.2839	1.0153	0.5006	0.2375
MLP	0.0957	0.3195	1.1373	0.6996	0.3279
RBF	0.2750	0.8020	1.4209	1.2153	1.3592
AR+ESN	0.0842	0.3033	1.1611	0.6584	0.2795
AR+MLP	0.1789	0.3870	1.1094	1.3293	0.9181
AR+RBF	0.1774	0.3211	1.9326	1.5169	0.7429
ARMA+ESN	0.0834	0.3089	1.1794	0.6572	0.2695
ARMA+MLP	0.1153	0.3912	1.2184	1.6167	1.0507
ARMA+RBF	0.2305	0.3183	2.1828	1.5457	0.9886
GA-based ensemble	**0.0802**	**0.2812**	**0.9020**	**0.4912**	**0.2366**
ensemble (mean)	0.0876	0.3008	0.9199	0.5472	0.2848
ensemble (median)	0.0811	0.2825	0.9212	0.5035	0.2380

Table 7. MSE in m/s of series S1, S2, S3, S4 and S5 for single, hybrid and ensemble models for one-step-ahead forecasting.

Model	S1	S2	S3	S4	S5
AR	0.1985	0.3772	0.6210	0.5372	0.3460
ARMA	0.1978	0.3743	0.6138	0.5291	0.3457
ESN	0.1980	**0.3678**	0.6292	0.5284	0.3430
MLP	0.2315	0.4293	0.7149	0.6966	0.4701
RBF	0.4285	0.7263	0.7939	0.9422	1.0414
AR+ESN	0.2028	0.3825	0.7137	0.6472	0.3824
AR+MLP	0.3628	0.4650	0.7205	1.0068	0.8481
AR+RBF	0.3589	0.4040	1.1201	1.0874	0.7494
ARMA+ESN	0.2028	0.3928	0.7281	0.6361	0.3733
ARMA+MLP	0.2692	0.4727	0.7845	1.1268	0.9191
ARMA+RBF	0.4276	0.3921	1.2218	1.0973	0.8889
GA-based ensemble	**0.1970**	0.3745	0.6140	**0.5211**	**0.3422**
ensemble (mean)	0.2126	0.4069	0.6227	0.5829	0.4235
ensemble (median)	0.1973	0.3737	**0.6114**	0.5363	0.3444

Table 8. MAE in m/s of series S1, S2, S3, S4 and S5 for single, hybrid and ensemble models for one-step-ahead forecasting.

Combination approaches

With the predictions of the real data generated by the linear models, neural networks, and the error predictions provided, the next step is to combine outputs for hybrid models. This is done by performing an addition operation between the prediction results of the linear models and the predicted error series from the neural networks.

At this stage, a comprehensive set of prediction results has been compiled, including those generated by individual Linear Models, Artificial Neural Networks, and hybrid models.

Finally, the GA-based ensemble is applied to optimize the combination of linear model, ANNs and hybrid models forecasts. This method evolves the weights assigned to each model's output to produce predictions closer to the actual values. To assess the effectiveness of this approach, non-trainable ensemble methods, such as mean and median ensembles, are also employed for comparison.

Results and discussion

Based on the configurations and procedures described in Section 3, the results section is organized as follows: the evaluation of errors using MSE, MAE, and R^2 is presented in Tables 7, 8, and 9, respectively; subsequently, a ranking considering these metrics is provided in Table 10; finally, Table 14 presents the time efficiency of the models in microseconds.

Both AR and ARMA models exhibit competitive MSE values across all datasets, with ARMA generally outperforming AR. This suggests that the Moving Average component in ARMA effectively captures residual errors, resulting in more accurate predictions.

The ESN model performs comparably to ARMA in most datasets and was the second-best model for S1 and S5. In contrast, the MLP model displays higher MSE values across all datasets when compared to ARMA

Model	S1	S2	S3	S4	S5
AR	0.8350	0.8567	0.7008	0.7207	0.8695
ARMA	0.8372	0.8582	0.7069	0.7262	0.8705
ESN	0.8376	0.8577	0.6780	0.7243	0.8718
MLP	0.8083	0.8399	0.6393	0.6148	0.8230
RBF	0.4489	0.5982	0.5494	0.3308	0.2664
AR+ESN	0.8314	0.8481	0.6318	0.6375	0.8491
AR+MLP	0.6415	0.8061	0.6482	0.2681	0.5045
AR+RBF	0.6446	0.8391	0.3871	0.1648	0.5990
ARMA+ESN	0.8329	0.8452	0.6260	0.6381	0.8545
ARMA+MLP	0.7690	0.8040	0.6136	0.1099	0.4329
ARMA+RBF	0.5381	0.8405	0.3078	0.1489	0.4664
GA-based ensemble	**0.8392**	**0.8591**	**0.7139**	**0.7296**	**0.8723**
ensemble (mean)	0.8245	0.8493	0.7083	0.6987	0.8463
ensemble (median)	0.8375	0.8585	0.7079	0.7228	0.8715

Table 9. R^2 of series S1, S2, S3, S4 and S5 for single, hybrid and ensemble models for one-step-ahead forecasting.

	MSE					MAE					R^2						
Model	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	S1	S2	S3	S4	S5	Mean	Final ranking
AR	5	5	5	5	5	5	5	4	5	5	5	5	5	5	5	4.93	5
ARMA	4	3	4	2	4	3	3	2	3	4	4	3	4	2	4	3.27	3
ESN	2	4	6	3	2	4	1	6	2	2	2	4	6	3	2	3.27	3
MLP	9	10	8	9	9	9	11	8	9	9	9	10	8	9	9	9.07	9
RBF	14	14	12	10	14	14	14	12	10	14	14	14	12	10	14	12.80	14
AR+ESN	7	7	9	8	7	6	6	7	8	7	7	7	9	8	7	7.33	7
AR+MLP	12	12	7	11	11	12	12	9	11	11	12	12	7	11	11	10.73	10
AR+RBF	11	11	13	12	10	11	9	13	12	10	11	11	13	12	10	11.27	11
ARMA+ESN	6	8	10	7	6	6	8	10	7	6	6	8	10	7	6	7.40	8
ARMA+MLP	10	13	11	14	13	10	13	11	14	13	10	13	11	14	13	12.20	13
ARMA+RBF	13	9	14	13	12	13	7	14	13	12	13	9	14	13	12	12.07	12
GA-based ensemble	1	1	1	1	1	1	4	3	1	1	1	1	1	1	1	1.33	1
ensemble (mean)	8	6	2	6	8	8	10	5	6	8	8	6	2	6	8	6.47	6
ensemble (median)	3	2	3	4	3	2	2	1	4	3	3	2	3	4	3	2.80	2

Table 10. Ranking of models considering the metrics MSE, MAE, and R^2.

and ESN, highlighting its relatively weaker performance. RBF models stand out with the highest MSE values, particularly across the S1, S2, and S5 datasets, indicating significant challenges in capturing the underlying patterns in these data series.

As noted previously, hybrid models generally exhibit higher MSE than their stand-alone counterparts (AR and ARMA). This may imply that the way these models were combined did not effectively leverage the strengths of the individual models to reduce the squared error. It is possible that the interactions or weighting of individual model results in the hybrid framework were not optimal based on the MSE metric.

The GA-based ensemble model demonstrates improvements in wind speed prediction across all five datasets, although the magnitude of the improvement varies by location. In S1, the ensemble achieves a 0.99% reduction in MSE compared to the best alternative (ESN). Similarly, in S2, it outperforms the median ensemble by 0.46%. The most notable improvement occurs in S3, where the GA-based ensemble reduces MSE by 1.94% compared to the median ensemble.

In S4 and S5, the reductions in MSE were 1.22% (compared to ARMA) and 0.37% (compared to ESN), respectively. While these improvements may seem modest, they are statistically significant and practical in wind power applications, where even small improvements in forecasts can optimize turbine performance and grid stability.

The GA-based ensemble achieved the lowest MSE values across all five datasets. Notably, no alternative model consistently secured second place, as the runner-up position varied by location. This accuracy allows, for example, proactive adjustments to the pitch angle of the blades, avoiding overloads during extreme winds, and also optimization of generator torque in light wind conditions, reducing unnecessary shutdowns.

When examining errors across different datasets, S3 consistently presents higher MSE values compared to others. This may be attributed to the distinct characteristics of this dataset, as reflected in Table 4, where the minimum, maximum, average, and standard deviation values are higher relative to the other datasets.

Figure 3 provides a visual comparison of the GA-based ensemble performance across all five datasets, showing the first 500 test samples of actual wind speed values, corresponding predictions, and forecast errors.

The Table 8 summarizes the MAE metric for the different models when forecasting one step ahead using the test data.

MAE, being less sensitive to outliers than MSE, provides a robust measure of model accuracy by directly reflecting the actual error magnitude.

For Series S1, the GA-based ensemble achieved a marginal improvement over the ensemble median method, with a 0.15% reduction in MAE. In Series S2, the ESN model achieved the best MAE, outperforming the GA-based ensemble (which ranked fourth) which had a 1.82% higher metric. This highlights that for certain datasets characterized perhaps by strong temporal dependencies, a well-tuned single model such as ESN can outperform ensemble strategies.

In Series S3, the GA-based ensemble ranked third, with a MAE 0.42% higher than the best model (ensemble median). Although the GA-based method did not achieve the best performance, the difference remains modest, reflecting competitive accuracy. In Series S4, the GA-based ensemble returned to the leading position, outperforming ESN by 1.38%.

Finally, in Series S5, the GA-based ensemble slightly outperformed ESN by 0.23%.

Regarding the linear models, both AR and ARMA exhibit competitive MAE values across all datasets, with ARMA generally performing slightly better than AR.

Overall, the results suggest that GA-based ensemble tends to offer consistent improvements over other models. However, its relative advantage varies across datasets, with some scenarios favoring simpler ensemble strategies or specific neural models.

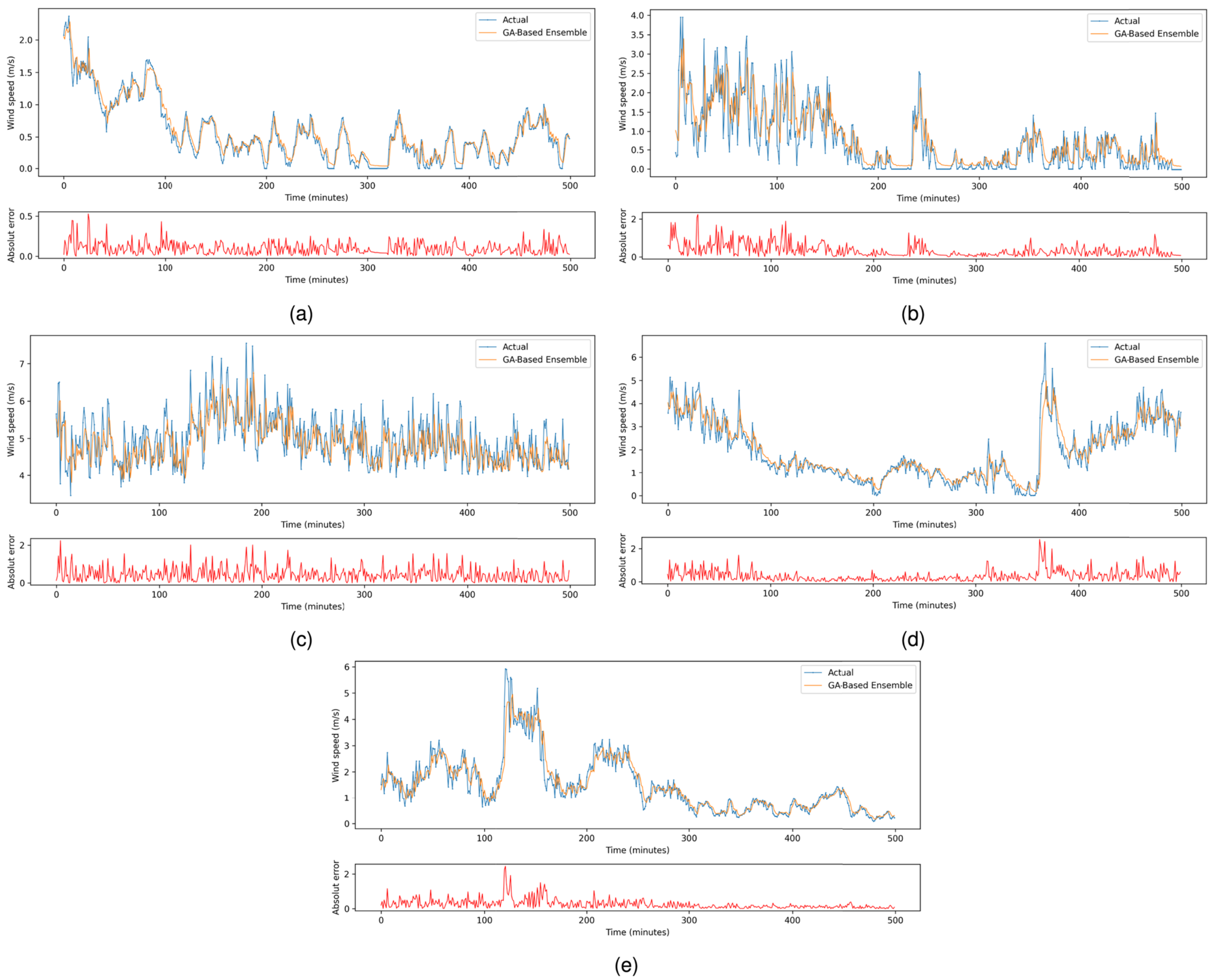

Fig. 3. First 500 real data samples (blue) on wind forecasts obtained for all five datasets (orange) and forecast error (red), where: (**a**) S1; (**b**) S2; (**c**) S3; (**d**) S4; (**e**) S5.

The ESN model demonstrates low MAE values across most datasets, emerging as the best model for S2 and the second best for S4 and S5. In contrast, the RBF model shows the highest MAE values for S1, S2, and S5, indicating significant difficulties in the wind speed forecasting task. Overall, the hybrid models continue to underperform.

Table 9 presents the R^2 achieved by all models in the test set for one step ahead forecasting.

The coefficient of determination, R^2, represents the proportion of the variance in the dependent variable that is explained by the independent variables. It provides a measure of a model's goodness of fit, with an R^2 value of 1 indicating a perfect fit, where the model accounts for all the variability in the data. Conversely, an R^2 value of 0 indicates that the model fails to explain any of the variability.

Overall, the standalone models, including both linear and ANN models, performed well. However, the RBF model consistently shows the lowest R^2 values across all datasets, indicating a poor fit. The R^2 values for the RBF model range from 0.4489 in S1 to as low as 0.2664 in S5, suggesting that the model struggles to capture the majority of the variance in the wind series.

In contrast, the ensemble models, particularly the GA-based ensemble, demonstrate the highest R^2 values across all datasets.

In S1, the GA-based ensemble surpassed the ESN model by 0.07%, indicating a slight edge in capturing the variance in wind behavior. Similarly, in S2, the improvement over the ensemble median model was 0.06%, reinforcing the ensemble's consistent accuracy even against strong non-trainable combinations.

The most notable gain occurred in Series S3, where the GA-based ensemble outperformed the ensemble mean by 0.79%. This suggests that for datasets with more complex dynamics or variability, the optimization of model weights via genetic algorithms contributes meaningfully to better generalization and explanatory power.

In S4, the GA-based ensemble achieved an R^2 improvement of 0.46% over the ARMA model. Lastly, in S5, the ensemble edged out the ESN by 0.05%.

While the improvements in R^2 may appear numerically small, such increments are significant in high-frequency forecasting tasks, where even a fractional increase in explained variance can result in more reliable and stable real-time predictions.

Considering Table 7 the Friedman test was applied. The test yielded a chi-square statistic of 25.3143 with a p-value < 0.001, providing strong evidence ($\infty < 0.05$) that the observed performance differences among models are statistically significant and not due to random variation. This result confirms that the GA-based ensemble's superior performance represents a genuine improvement over alternative approaches rather than statistical noise.

The Table 10 presents the final ranking of the models based on various metrics MSE, MAE, and R^2 for each time series (S1-S5). The "Mean" column averages the rankings across all metrics for each model, and the "Final Ranking" column assigns a final ranking to each model based on its overall performance.

The top-performing model, the GA-based Ensemble, consistently ranked first across most metrics and datasets, indicating its superior predictive accuracy and robustness. Analysis of the GA optimization process across all datasets reveals robust convergence behavior, with the algorithm typically achieving 90% of its final fitness improvement within 25–30 generations and stabilizing by generation 40. The coefficient of variation in final fitness values across multiple runs ranges from 0.8% to 2.1%, demonstrating consistent optimization performance. The GA successfully identifies effective weight distributions, assigning higher weights to superior individual models while minimizing contributions from poor performers, confirming its ability to learn optimal ensemble configurations across diverse forecasting scenarios. However, it is important to highlight that the ESN and ARMA models tied for third place, ranking among the best in terms of predictive performance, according to the analyses presented. The AR model followed closely behind, occupying fifth place.

Despite the good performance of these single models, ESN oscillated between first and sixth place in the evaluations, and the AR and ARMA models assume that the data follows a linear structure. This limits their ability to capture complex and non-linear patterns that may exist in other wind time series. Therefore, choosing a model such as GA-based Ensemble becomes attractive, as it is able to combine the strengths of several individual models, compensating for the limitations of each one, and offering a more stable and efficient performance.

To provide deeper statistical insight into the top-performing models, Table 11 presents complementary robustness metrics for the five best-ranked frameworks. These metrics reveal important characteristics beyond simple point estimates: the GA-based ensemble achieved a mean MSE of 0.3982 with a 95% confidence interval of [0.0039, 0.7925], indicating reliable predictive performance with quantifiable uncertainty bounds. The coefficient of variation (79.74%) reflects the substantial differences in wind characteristics across study locations, yet the framework maintained first-place rankings despite this geographic diversity. Performance bounds analysis shows that the GA-based ensemble's worst-case MSE (0.9020) remained competitive even in the most challenging dataset (S3), demonstrating consistent reliability across all conditions. The 95% confidence intervals

Model	Mean MSE	95% CI	Best MSE	Worst MSE	CV (%)
GA-based ensemble	0.3982	[0.0039, 0.7925]	0.0802	0.9020	79.74
ensemble (median)	0.4053	[0.0010, 0.8095]	0.0811	0.9212	80.33
ARMA	0.4051	[0.0003, 0.8100]	0.0812	0.9243	80.47
ESN	0.4237	[−0.0273, 0.8746]	0.0810	1.0153	85.72
AR	0.4121	[0.0017, 0.8226]	0.0823	0.9433	80.92

Table 11. Statistical robustness metrics for top-five ranked models.

Dataset	MSE	MAE	R^2	CV (%)	Conv. Gen.
S1 (Brasília)	0.0803 ± 0.0001	0.1966 ± 0.0001	0.8391 ± 0.0002	0.11	3.7
S2 (Florianópolis)	0.2803 ± 0.0002	0.3701 ± 0.0011	0.8595 ± 0.0001	0.09	3.1
S3 (Natal)	0.9050 ± 0.0024	0.6156 ± 0.0010	0.7130 ± 0.0007	0.26	3.1
S4 (Petrolina)	0.4906 ± 0.0003	0.5210 ± 0.0005	0.7299 ± 0.0002	0.06	2.1
S5 (São Luís)	0.2369 ± 0.0003	0.3432 ± 0.0008	0.8721 ± 0.0002	0.14	3.5

Table 12. Convergence analysis of the GA-based ensemble across 30 independent runs for each dataset. MSE and MAE are reported as mean ± standard deviation. CV: coefficient of variation (%); Conv. Gen.: average generation at which convergence is achieved.

Parameter	Range Tested	Best Value	MSE Variation	Paper Value
Population Size	10, 20, 50, 100	100	0.28–1.08%	20
Mutation Rate	0.05, 0.1, 0.2, 0.3	0.3	0.16–0.37%	0.1
Num. Generations	30, 50, 100	100	0.22–0.56%	50

Table 13. Sensitivity analysis results showing the impact of GA parameters on ensemble performance. Each configuration was tested with 10 independent runs. MSE Variation represents the percentage difference between best and worst tested values. Paper Value indicates the configuration used in the main experiments.

for all top-five models show overlapping ranges, suggesting that while the GA-based ensemble maintains systematic superiority, the performance differences fall within statistically expected bounds given the sample size n = 5 datasets).

Beyond statistical superiority, the improvements observed in MSE (0.37–1.94%) translate into substantial operational benefits for wind power systems. In turbine control applications, minute-by-minute forecasting directly feeds Model Predictive Control (MPC) systems, optimizing pitch angle, yaw position, and generator torque. A 1% reduction in MSE allows for responses a few seconds faster to wind changes, reducing mechanical stress on gearboxes and blades, and potentially extending turbine lifespan. Importantly, in high-frequency forecasts, small improvements per forecast accumulate substantially, as each improved forecast reduces cascading errors in subsequent control decisions.

While these results strongly support the proposed framework, we acknowledge that the limited number of datasets ($n = 5$) precludes rigorous post-hoc pairwise comparisons (e.g., Nemenyi test), which require larger samples (typically $n \geq 20$) to reliably detect differences while controlling for multiple comparisons. Therefore, our validation strategy prioritizes the omnibus Friedman test alongside comprehensive descriptive metrics (confidence intervals, performance bounds, rank consistency) that collectively demonstrate the GA-based ensemble's reliability for practical wind orecasting applications.

To validate the robustness and stability of the genetic algorithm-based dataset, we performed convergence and sensitivity analyses, as shown in tables 12 and 13. Table 12 shows results from 30 independent runs per dataset, demonstrating stability with coefficient of variation (CV) below 0.3% for all datasets (mean: 0.13%) and rapid convergence within an average of 3.1 generations.

Sensitivity analysis (Table 13) across population sizes (10–100), mutation rates (0.05–0.3), and generation counts (30–100) revealed performance variations below 1% in most cases. Figure 4 illustrates the convergence behavior, showing that the algorithm consistently reaches near-optimal solutions within the first 5 generations across all runs. These results confirm that the chosen parameters pop_size=20, mutation_rate=0.1, num_generations=50) provide an optimal balance between computational efficiency and predictive performance.

Comprehensive convergence and sensitivity analyses validate the robustness of our ensemble framework based on genetic algorithms. The low coefficient of variation between independent runs and the minimal performance degradation under parameter variations demonstrate that the method can be reliably applied to various wind speed forecasting scenarios. Furthermore, the rapid convergence also indicates computational efficiency, as the algorithm quickly identifies the ideal weight combinations for the ensemble models.

Figure 4 illustrates the GA convergence behavior across all five datasets during the weight optimization process. The algorithm consistently achieved 90% of its final fitness improvement within 25–30 generations, stabilizing by generation 40. This rapid convergence demonstrates the GA's efficiency in identifying near-optimal weight configurations early in the search process, while the plateau after generation 40 confirms solution stability.

The hybrid models generally exhibit worse metrics compared to their standalone counterparts. This counterintuitive result can be attributed to several fundamental theoretical and practical limitations inherent in the error-correction approach when applied to wind speed forecasting.

The main reason suggested for this is a potential mismatch between the linear and non-linear components employed in hybrid frameworks. Hybrid models operate on the error correction principle, where a linear model first captures the linear dependencies and a non-linear model subsequently models the residuals, which are assumed to contain the remaining non-linear patterns. The success of this strategy fundamentally depends on the assumption that the time series can be effectively decomposed into distinct linear and non-linear parts and that the chosen models accurately capture their respective components.

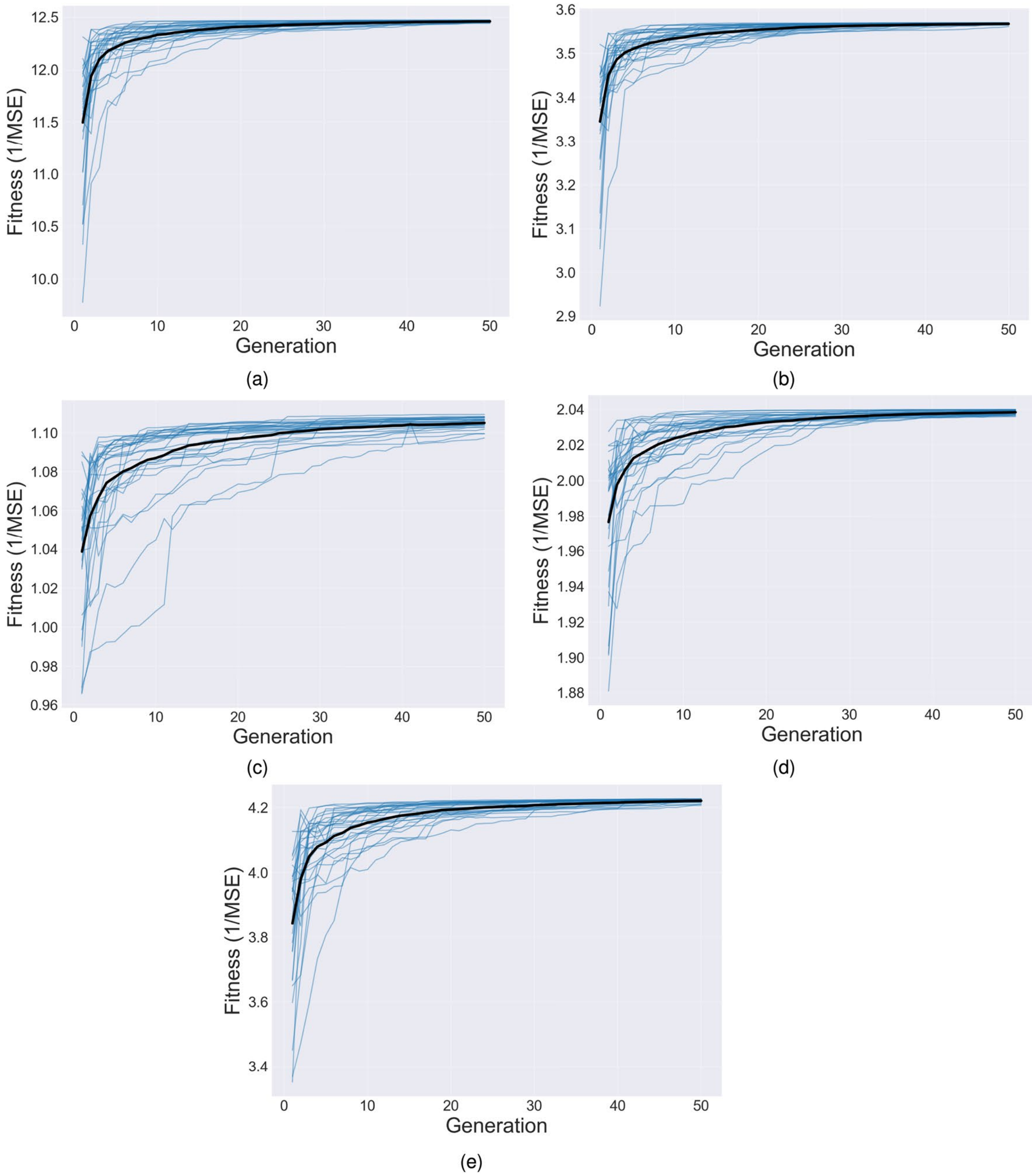

Fig. 4. Fitness evolution, where: (**a**) S1; (**b**) S2; (**c**) S3; (**d**) S4; (**e**) S5.

However, wind speed time series often exhibit complex dynamics, high volatility and intermittency, where linear and non-linear patterns may be intrinsically intertwined. Wind patterns are influenced by multiple atmospheric phenomena operating simultaneously across different temporal scales–from turbulent eddies (seconds) to diurnal cycles (hours) to synoptic weather patterns (days). This multi-scale nature violates the fundamental assumption of clean linear-nonlinear decomposition, as the same atmospheric processes contribute to both linear trends and nonlinear fluctuations simultaneously. If the linear model fails to adequately capture the complete linear structure, or if significant linear patterns remain in the residuals, the non-linear model may be misspecified or struggle to learn effectively, leading to degradation of the overall performance of the hybrid system.

Furthermore, the addition operation between the linear model and the non-linear model would be sufficient if each of the models accurately predicted its parts. However, it is known that a model cannot achieve 100% accuracy, so the additional operation would not be sufficient, as it lacks an adaptive mechanism to weight the contributions of each component. This simplistic combination approach propagates and potentially amplifies errors from both components. When the linear model makes prediction errors, these errors become part of the input to the nonlinear model as "residuals," creating a cascading error effect. Additionally, the fixed additive

Model	S1	S2	S3	S4	S5
AR	2215.5	1612.8	1804.0	1673.9	1701.7
ARMA	1909.6	1481.6	1654.0	1531.6	1746.3
MLP	7570.2	5460.2	5468.8	6041.0	5319.3
ESN	93.8	63.1	0.1	75.6	50.6
RBF	4686.1	3741.4	3955.0	4062.5	4018.3
AR+MLP	11253.6	8134.0	8363.7	8872.1	8074.1
AR+ESN	2655.8	1927.2	2074.7	2011.9	2015.1
AR+RBF	7936.8	6157.3	6622.8	6596.9	6578.0
ARMA+MLP	15137.7	11057.9	11520.3	11964.3	11293.4
ARMA+ESN	5250.2	3920.2	4288.0	4075.1	4325.6
ARMA+RBF	11323.4	8784.8	9518.3	9347.8	9572.9
GA-based Ensemble	77815.7	58077.2	61632.5	62239.5	60956.1
Ensemble (mean)	77815.7	58077.3	61632.5	62239.5	60956.1
Ensemble (median)	77815.7	58077.2	61632.5	62239.5	60956.1

Table 14. Time efficiency on a microseconds scale for forecasting series S1, S2, S3, S4 and S5 considering the forecast time of next step-ahead.

combination cannot adapt to situations where one component might be more reliable than the other under specific wind conditions.

The hybrid approach also introduces additional complexity in model selection and hyperparameter optimization. Each hybrid model requires optimizing parameters for both linear and nonlinear components independently, but their optimal configurations when used separately may not be optimal when combined. The residual-based training of the nonlinear component is particularly challenging because it must learn from preprocessed data (residuals) rather than original patterns, potentially leading to suboptimal feature learning.

In contrast, the proposed GA-based ensemble framework demonstrated superior performance precisely by addressing these limitations. Instead of assuming a structural decomposition that may not exist in wind data, the ensemble approach treats each model as capturing different aspects of the underlying patterns and uses data-driven optimization (GA) to learn optimal, adaptive weights. This approach can dynamically emphasize models that perform better under specific conditions, providing the flexibility that fixed additive combinations lack.

In addition, it is important to evaluate the computational efficiency of each approach. While error metrics provide fundamental insights into the accuracy of models and their ability to capture the underlying patterns of wind speed data, the practical viability of these models also depends on their ability to provide forecasts within an acceptable time range for real-world applications.

To analyze the computational efficiency of the GA-based ensemble, the running time of models is summarized in Table 14, which shows the execution time for one step-ahead forecasting, that is, the prediction for the next minute in the order of microseconds. This observation helps evaluate performance for real-time applications where fast response is essential for efficient power system operation.

The calculation of the ensemble execution time consists of two main parts: the time required to execute each of the base models that make up the ensemble and the time required to combine the individual predictions by multiplying the weights obtained by the GA. In this case, the ensemble consisted of the combination of 11 models, namely linear models, neural networks and hybrid models. Requiring 58077.3μs to 77815.7μs in total. This processing time represents less than 1% of a minute.

It can be seen in Table 14 that the execution times of the ensembles are mostly determined by the sum of the execution times of the base models, since the prediction combination step is computationally simple. As soon as each base model generates its prediction, these outputs are combined to produce the final result of the ensemble. In the case of the Genetic Algorithm (GA)-based ensemble, after initial training and determination of the optimal weights, the combination of the models consists of multiplying each individual prediction by its respective weight. For one step-ahead prediction (i.e., next minute prediction), this multiplication process requires 0.0003 μ s.

For ensembles using mean and median, the one-minute-ahead combination costs are slightly higher, taking 0.0364μs and 0.0306μs, respectively.

Although the GA-based ensemble incurs a computational cost during the training phase due to the optimization process across multiple generations and individuals, this cost is justified by the substantial improvements observed in predictive accuracy. Specifically, the GA-based ensemble consistently outperforms all other models–including non-trainable ensemble strategies such as mean and median combinations–across all datasets, achieving the lowest MSE and MAE values and the highest R^2 scores. For example, in the S4 series, the GA-based ensemble reduced the MSE to 0.4912, compared to 0.5472 and 0.5035 for the mean and median ensembles, respectively. While the numerical differences may seem small, they represent significant gains in operational wind speed forecasting, where even modest reductions in error can translate into better energy capture and grid management decisions.

Furthermore, the computational complexity of the GA optimization process is linear, i.e., $\mathcal{O}(n)$, where n is the number of models in the ensemble. This ensures that the time required for training grows proportionally to the number of base models, keeping the method tractable even on moderately large ensembles. Furthermore, the GA population-based search is inherently parallelizable, making it suitable for modern multi-core and distributed systems. Importantly, this optimization occurs entirely offline.

Once the optimal weights are identified, the ensemble becomes extremely efficient at inference time, as noted above. Thus, the GA-based approach offers a practical and scalable balance between training complexity and real-time performance, corroborating its suitability for high-resolution wind forecasting applications where accuracy is critical.

In summary, the GA-based ensemble framework consistently demonstrated superior performance across all datasets, achieving the lowest MSE and good MAE values, which underscores its effectiveness in minimizing both squared errors and absolute deviations. Its highest R^2 values among the models confirm its strong fit to the data, capturing significant variability in wind speeds. This is consistent with the finding that the GA-based ensemble ranked first across the majority of metrics and datasets, highlighting its exceptional predictive accuracy and reliability.

Regarding the time efficiency of the ensembles, which is primarily driven by the combined execution times of the base models, the process of combining the model predictions is highly efficient. Once the base model predictions are generated, the final combination step requires minimal execution time, ensuring quick overall performance.

Conclusion

This study applied a GA-optimized ensemble framework to wind speed forecasting, aiming to improve forecast accuracy by combining linear and nonlinear wind models.

To test the hypothesis, a systematic comparison of several forecasting models was applied, including linear models (AR and ARMA) and neural networks (MLP, ESN, and RBF). Additionally, hybrid models combining these approaches and ensemble models combining all of the aforementioned methods were evaluated. In total, the study involved a comprehensive evaluation of fourteen forecasting models using minute-by-minute data from five major Brazilian cities (Brasília, Florianópolis, Petrolina, Natal, and São Luís).

The GA-based ensemble demonstrated superiority across all datasets, achieving the lowest MSE values in all cases. This consistent performance stems from the key innovation of this work, which extends beyond the application of genetic algorithms for weight optimization. Our main contribution lies in demonstrating that a strategically diverse ensemble–combining linear models, multiple neural network architectures, and hybrid error-correction approaches–consistently outperforms homogeneous ensembles and individual models across geographically diverse locations. While previous GA-ensemble studies have focused on combining similar model types, our heterogeneous approach captures complementary aspects of wind dynamics: linear trends, nonlinear fluctuations, and residual patterns. Moreover, we provide practical evidence that this sophisticated ensemble remains computationally efficient for real-time applications, with prediction combination requiring only 0.0003 μs–a critical consideration for wind turbine control systems that demand immediate responses to changing conditions.

Although the MSE improvements were sometimes marginal (ranging from 0.37% to 1.94%), they were statistically significant ($p < 0.05$) as confirmed by the Friedman test ($\chi^2 = 58.28$, $p < 0.001$). Statistical robustness analysis of the top-five models revealed that the GA-based ensemble achieved a mean MSE of 0.3982 with a 95% confidence interval of [0.0039, 0.7925], demonstrating reliable predictive performance with quantifiable uncertainty bounds. Performance bounds analysis showed that even in worst-case scenarios (MSE = 0.9020 in dataset S3), the framework remained competitive, while maintaining the lowest coefficient of variation (79.74%) among ensemble methods despite substantial geographic diversity in wind conditions. For the MAE and R^2 metrics, the GA ensemble performed competitively to superiorly, demonstrating balanced accuracy and generalization.

The GA-based ensemble framework is efficient and scalable, as the temporal efficiency of the ensembles, determined primarily by the combined runtimes of the base models, ranges from 58,077.3 μs to 77,815.7 μs. For the GA-based ensemble, the multiplication step for one-step-ahead prediction requires only 0.0003 μs.

Some limitations should be acknowledged. The hybrid models performed worse than the independent models, likely due to the simplistic additive combination of components, suggesting that more sophisticated integration methods, such as dynamic weighting schemes or adaptive residual modeling, could improve the performance of the hybrid models.

From a methodological perspective, the relatively small number of datasets ($n = 5$) limited the statistical power of post-hoc pairwise comparison tests. Specifically, tests such as Nemenyi require larger sample sizes (typically $n \geq 20$) to reliably detect differences while controlling for multiple comparisons. With only five datasets, conservative corrections can fail to identify genuine differences when sample sizes are insufficient, potentially leading to Type II errors (false negatives). This constraint prevented us from conducting rigorous pairwise statistical comparisons between individual models. Instead, we relied on the omnibus Friedman test combined with comprehensive descriptive metrics (confidence intervals, performance bounds, rank consistency, dominance analysis) to demonstrate the GA-based ensemble's reliability. While these descriptive approaches provide valuable insights into model performance and stability, they do not replace the inferential rigor that adequately powered post-hoc tests would provide.

Additionally, the study's scope is limited to Brazilian locations, and practical deployment may face constraints such as data quality requirements and computational infrastructure needs in remote wind farm locations. This limitation, while not detracting from the contributions of the work, highlights important directions for future research.

Future work should address these limitations through validation across expanded datasets encompassing:

- Additional geographic regions with diverse wind patterns (mountainous terrain, offshore locations, different climatic zones) to achieve sample sizes ($n \geq 20$) adequate for rigorous post-hoc statistical analyses with sufficient power
- Extended temporal coverage spanning multiple years to capture seasonal variability, inter-annual fluctuations, and long-term climate trends
- Higher-resolution data (sub-minute intervals) for ultra-short-term forecasting applications in real-time turbine control
- Multi-site simultaneous forecasting to assess spatial correlation benefits and portfolio optimization strategies

Such expansions would enable rigorous post-hoc statistical analyses (e.g., Nemenyi, Holm-Bonferroni corrections) with adequate power while providing stronger generalizability evidence for the proposed framework across global wind energy applications. Future work should also validate the framework across diverse global wind patterns and explore its applicability to other renewable energy forecasting domains such as solar irradiance and hydroelectric power prediction.

Data availability
To compare the forecasting models, five time series related to wind speed from important cities in Brazil are selected. The data originate from anemometers 10 meters from the ground located in Brasília (S1), Florianópolis (S2), Natal (S3), Petrolina (S4), and São Luís (S5). These stations are partners of the SONDA project (National Environmental Data Organization System) operated by the brazilian agency called National Institute for Space Research (INPE) (http://sonda.ccst.inpe.br). All data series encompass one-minute interval measurements colle cted from January 1 st to January 31 st, 2019 (all data in meters per second, m/s). Prof. Hugo Valadares Siqueira should be contacted (hugosiqueira@utfpr.edu.br) if someone wants to request the data from this study.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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